

# On Rupert's problem<sup>1</sup>

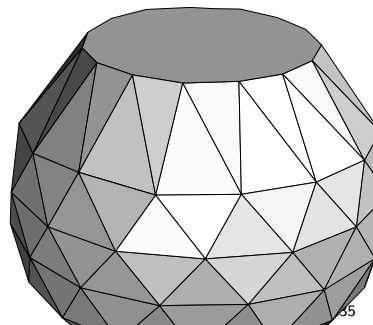
PolSys Seminar (Sorbonne University)

Sergey Yurkevich

17<sup>th</sup> April, 2026

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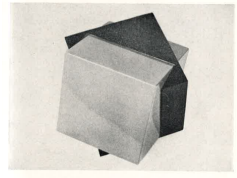
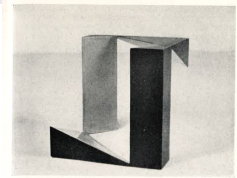
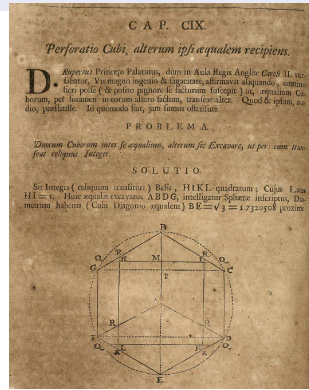
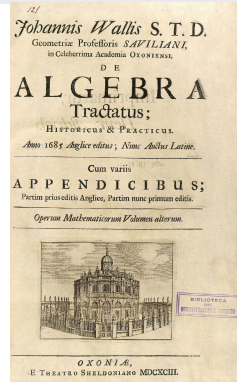
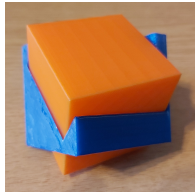
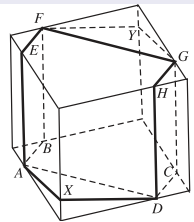
<sup>1</sup>Based on joint work with Jakob Steinerger:  
[arxiv.org/abs/2508.18475](https://arxiv.org/abs/2508.18475) and [arxiv.org/abs/2112.13754](https://arxiv.org/abs/2112.13754)



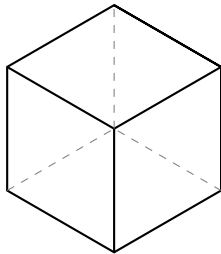
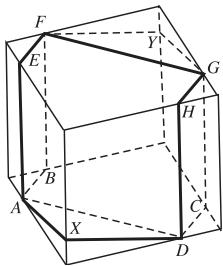
# Prince Rupert's cube

## Fact (Wallis, 1685)

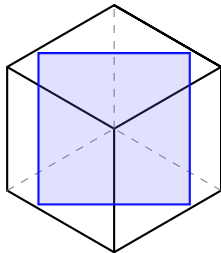
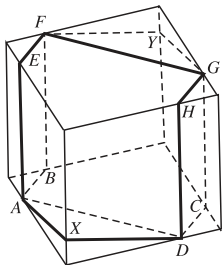
It is possible to cut a hole inside the **unit cube** such that another **unit cube** can pass through this hole.



# Definition of Rupert's problem



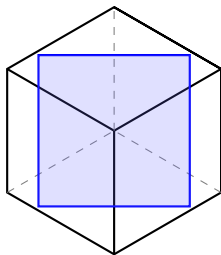
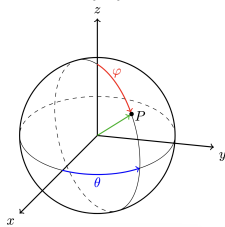
# Definition of Rupert's problem





# Definition of Rupert's problem

Let  $M(\theta, \varphi) : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be an orthogonal projection map in direction  $X(\theta, \varphi) \in \mathbb{R}^3$  and  $R(\alpha) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the rotation map.



$$X(\theta, \varphi) := (\cos \theta \sin \varphi, \sin \theta \sin \varphi, \cos \varphi)^t,$$

$$M(\theta, \varphi) := \begin{pmatrix} -\sin(\theta) & \cos(\theta) & 0 \\ -\cos(\theta) \cos(\varphi) & -\sin(\theta) \cos(\varphi) & \sin(\varphi) \end{pmatrix},$$

$$R(\alpha) := \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}.$$

## Definition

A point-symmetric polyhedron  $\mathbf{P}$  has Rupert's property, if there exist 5 parameters  $\alpha, \theta_1, \theta_2 \in [0, 2\pi)$  and  $\varphi_1, \varphi_2 \in [0, \pi]$  such that

$$R(\alpha) \circ M(\theta_1, \varphi_1) \mathbf{P} \subset (M(\theta_2, \varphi_2) \mathbf{P})^\circ.$$

## Brief history of Rupert's problem

- The Cube is Rupert [conjectured by Prince Rupert, proved by Wallis 1685].
- The “optimal” solution for the Cube [Nieuwland, 1816].

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Conjecture (Jerrard, Wetzel, Yuan, 2017 and Chai, Yuan, Zamfirescu, 2018)

*All convex polyhedra have Rupert's property.*

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- 9 of 13 Archimedean solids are Rupert [Hoffmann, 2018] [Lavau, 2019].

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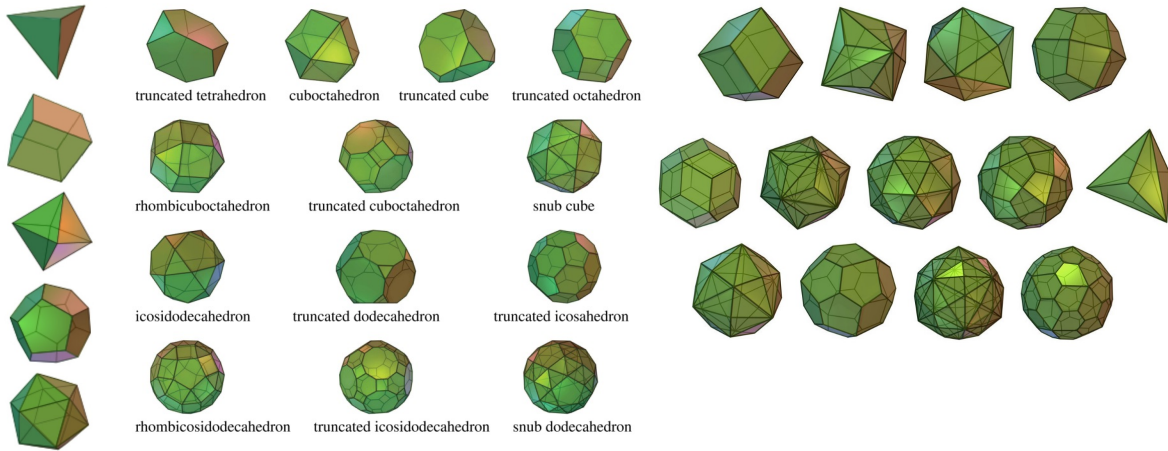
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- 9 of 13 Archimedean solids are Rupert [Hoffmann, 2018] [Lavau, 2019].
- 10 of 13 Archimedean solids and many other polyhedra are Rupert. Efficient way for proving Rupert's property. Theoretical algorithm for deciding. [S., Y., 2021].
- 11 of 13 Catalan solids are Rupert, improved optimization [Fredriksson, 2022].
- The **Noperthedron**: a counter example to the conjecture [Steininger, Y., 2025].



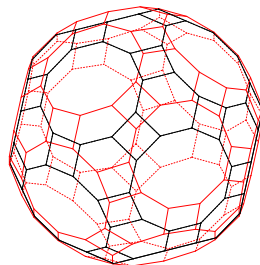
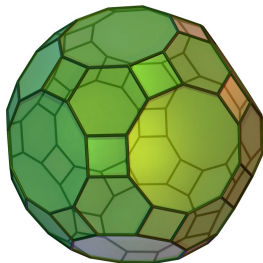
# Solids



# The Truncated Icosidodecahedron

Theorem (Steininger, Y., 2021)

*The Truncated Icosidodecahedron has Rupert's property.*



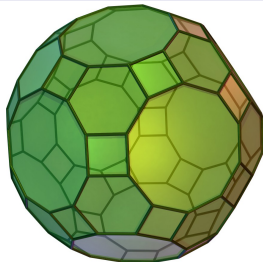
# The Truncated Icosidodecahedron

Theorem (Steininger, Y., 2021)

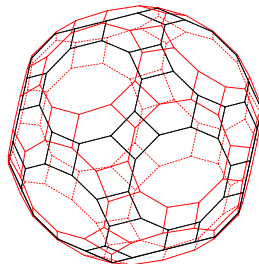
*The Truncated Icosidodecahedron has Rupert's property.*

Proof.

$\alpha = 0.43584, \theta_1 = 2.77685, \theta_2 = 0.79061, \varphi_1 = 2.09416, \varphi_2 = 2.89674$ ,  
plus some verification of linear inequalities in Maple/SageMath. □



$$R(\alpha) \circ M(\theta_1, \varphi_1) \mathbf{P} \subset (M(\theta_2, \varphi_2) \mathbf{P})^\circ.$$

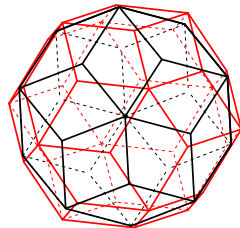
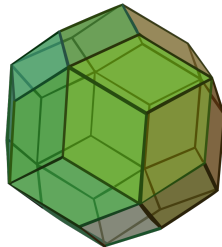
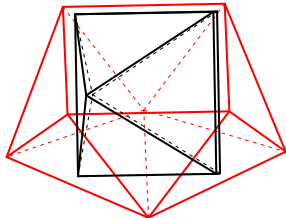


# Platonic, Archimedean, Catalan and Johnson Solids

## Theorem (Steininger, Y., 2021)

*In a few minutes it can be automatically proven that:*

- 1** *All 5 Platonic solids are Rupert.*
- 2** *At least 10 out of 13 Archimedean solids are Rupert.*
- 3** *At least 9 out of 13 Catalan solids are Rupert.*
- 4** *At least 82 out of 92 Johnson solids are Rupert.*



*Let  $\mathbf{P}$  be a convex polyhedron with algebraic coordinates. There exists a deterministic algorithm deciding whether  $\mathbf{P}$  is Rupert and finding a solution if it exists.*

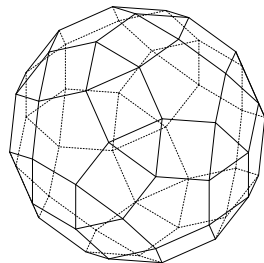
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# Are all convex polyhedra Rupert?

Conjecture (Jerrard, Wetzel, Yuan, 2017 and Chai, Yuan, Zamfirescu, 2018)

*All convex polyhedra have Rupert's property.*

- All Platonic solids are Rupert.
- 3 Archimedean solids remain open. One of them is point-symmetric.

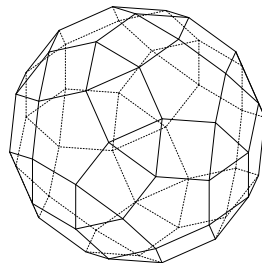


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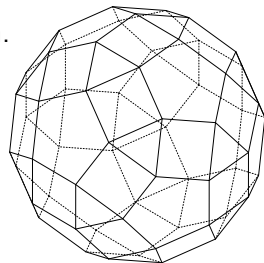


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- *Rupertness*: Probability that a random projection yields a solution.
- Can estimate confidence intervals for Rupertness.
- Conclusion: RID is significantly different from other regular solids. It is likely that RID is a counterexample to the conjecture.

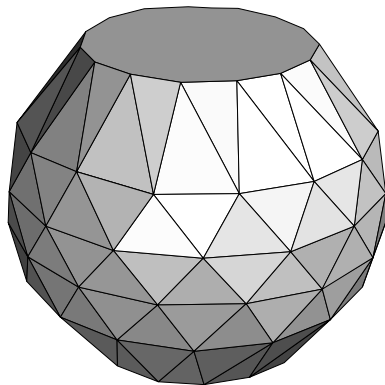




## 4 years later: The Noperthedron is not Rupert

Theorem (Steininger, Y., 2025)

*The Noperthedron, **NOP**, does not have Rupert's property.*



# Idea of proof

- Partition the five-dimensional solution space

$$I = [0, 2\pi) \times [0, \pi] \times [0, 2\pi) \times [0, \pi] \times [-\pi, \pi)$$

into small parts and prove for each that no solution in that region exists.

- Rough idea: Show that the “middle point” of any region does not yield a solution and argue with effective continuity of the parameters that this also excludes an explicit neighborhood around that point.

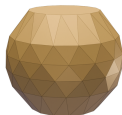
# Idea of proof

- Partition the five-dimensional solution space

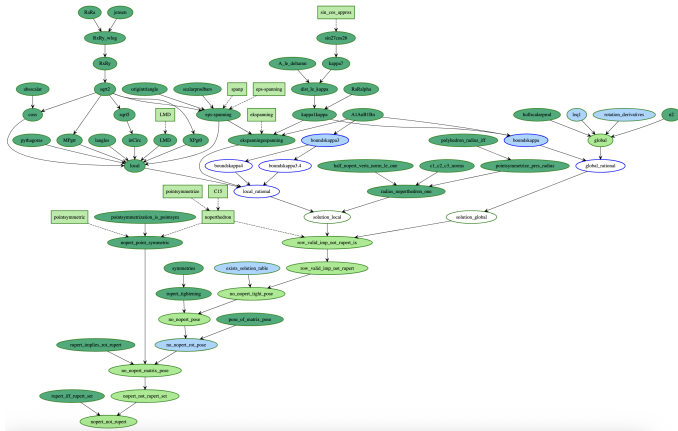
$$I = [0, 2\pi) \times [0, \pi] \times [0, 2\pi) \times [0, \pi] \times [-\pi, \pi)$$

into small parts and prove for each that no solution in that region exists.

- Rough idea: Show that the “middle point” of any region does not yield a solution and argue with effective continuity of the parameters that this also excludes an explicit neighborhood around that point.
- The **global theorem** is tailored for two generic projections of  $\mathbf{P}$ , when some vertex of the “smaller” projection  $\mathcal{P} = R(\alpha)M(\theta_1, \varphi_1)\mathbf{P}$  is *strictly outside* the “larger” projection  $\mathcal{Q} = M(\theta_2, \varphi_2)\mathbf{P}$ .
- The **local theorem** can handle projections that look (almost) exactly the same, for instance if  $\theta_1 \approx \theta_2$ ,  $\varphi_1 \approx \varphi_2$  and  $\alpha \approx 0$ .

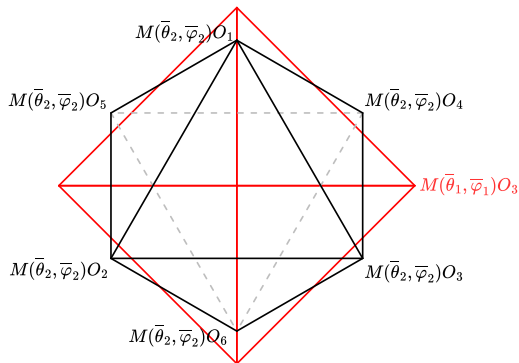


- [Blueprint](#)
- [Blueprint as pdf](#)
- [Dependency graph](#)
- [Doc pages for this repository](#)



Community **Project** formalizing the proof in LEAN by J. Reed and D. Renshaw.

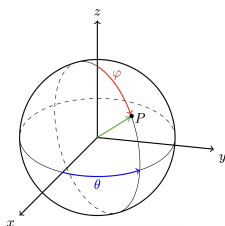
# Motivation for global theorem



$$(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) = (0, 0, \pi/4, \tan^{-1}(\sqrt{2}), 0)$$

# Starting point of the global theorem

Recall:



$$X(\theta, \varphi) := (\cos \theta \sin \varphi, \sin \theta \sin \varphi, \cos \varphi)^t,$$

$$M(\theta, \varphi) := \begin{pmatrix} -\sin(\theta) & \cos(\theta) & 0 \\ -\cos(\theta) \cos(\varphi) & -\sin(\theta) \cos(\varphi) & \sin(\varphi) \end{pmatrix},$$

$$R(\alpha) := \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}.$$

## Lemma

Let  $\varepsilon > 0$  and  $|\theta - \bar{\theta}|, |\varphi - \bar{\varphi}|, |\alpha - \bar{\alpha}| \leq \varepsilon$ , then

- $\|X(\theta, \varphi) - X(\bar{\theta}, \bar{\varphi})\| < \sqrt{2}\varepsilon,$
- $\|M(\theta, \varphi) - M(\bar{\theta}, \bar{\varphi})\| < \sqrt{2}\varepsilon,$
- $\|R(\alpha)M(\theta, \varphi) - R(\bar{\alpha})M(\bar{\theta}, \bar{\varphi})\| < \sqrt{5}\varepsilon.$

# First version of the global theorem

Recall the definition of Rupert's property

$$R(\alpha)M(\theta_1, \varphi_1) \mathbf{P} \subset (M(\theta_2, \varphi_2) \mathbf{P})^\circ.$$

If  $\Psi \in \mathbb{R}^5$  is a solution, then for any vector  $w \in \mathbb{R}^2$  it holds that

$$\langle R(\alpha)M(\theta_1, \varphi_1)S, w \rangle < \max_{P \in \mathbf{P}} \langle M(\theta_2, \varphi_2)P, w \rangle.$$

## Theorem (Global theorem v0.1)

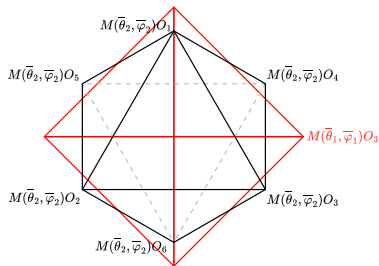
Let  $\mathbf{P}$  be convex, pointsymmetric with radius 1. Assume:

$$\langle R(\bar{\alpha})M(\bar{\theta}_1, \bar{\varphi}_1)S, w \rangle > \max_{P \in \mathbf{P}} \langle M(\bar{\theta}_2, \bar{\varphi}_2)P, w \rangle + (\sqrt{2} + \sqrt{5})\varepsilon$$

for some  $S \in \mathbf{P}$  and  $w \in \mathbb{R}^2$  with  $\|w\| = 1$ , then there cannot be a solution  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta}_1 \pm \varepsilon, \bar{\varphi}_1 \pm \varepsilon, \bar{\theta}_2 \pm \varepsilon, \bar{\varphi}_2 \pm \varepsilon, \bar{\alpha} \pm \varepsilon]$ .

# Example

- Octahedron  $\mathbf{O} = \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\} \subseteq \mathbb{R}^3$  and projection directions  $(\bar{\theta}_1, \bar{\varphi}_1) = (0, 0)$  and  $(\bar{\theta}_2, \bar{\varphi}_2) = (\pi/4, \tan^{-1}(\sqrt{2}))$ . Set  $\bar{\alpha} = 0$ .
- Goal: show this is not a solution to Rupert's problem and there exists  $\varepsilon > 0$  such that there is also no solution  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha)$  with  $|\bar{\theta}_i - \theta_i|, |\bar{\varphi}_i - \varphi_i|, |\alpha| \leq \varepsilon$ .





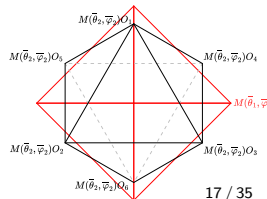
## Example continued

Following the global theorem we first compute:

$$M(\bar{\theta}_1, \bar{\varphi}_1) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad \text{and} \quad M(\bar{\theta}_2, \bar{\varphi}_2) = \begin{pmatrix} -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{6}/6 & -\sqrt{6}/6 & \sqrt{6}/3 \end{pmatrix}.$$

We choose  $S = O_3 = (0, 1, 0)$  and  $w = (1, 0)$ , thus  $\langle R(\bar{\alpha})M(\bar{\theta}_1, \bar{\varphi}_1)S, w \rangle = 1$ .

$$\langle M(\bar{\theta}_2, \bar{\varphi}_2)P, w \rangle = \begin{cases} 0 & \text{for } P = M(\bar{\theta}_2, \bar{\varphi}_2)O_i \text{ with } i = 1, 6, \\ \sqrt{2}/2 & \text{for } P = M(\bar{\theta}_2, \bar{\varphi}_2)O_i \text{ with } i = 3, 4, \\ -\sqrt{2}/2 & \text{for } P = M(\bar{\theta}_2, \bar{\varphi}_2)O_i \text{ with } i = 2, 5. \end{cases}$$



## Example continued

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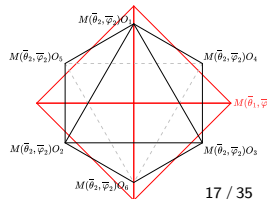
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Thus if  $\varepsilon > 0$  is chosen so that

$$1 > \sqrt{2}/2 + (\sqrt{2} + \sqrt{5})\varepsilon, \quad \text{e.g., } \varepsilon = 0.08$$

there is no solution in

$$(0, 0, \pi/4, \tan^{-1}(\sqrt{2}), 0) \pm \varepsilon \subseteq \mathbb{R}^5.$$



# Actual Global theorem

## Theorem (Global Theorem v1.0)

Let  $\mathbf{P}$  be a pointsymmetric convex polyhedron with radius 1 and let  $S \in \mathbf{P}$ . Let  $w \in \mathbb{R}^2$  be a unit vector and denote  $\overline{M}_1 := M(\overline{\theta}_1, \overline{\varphi}_1)$ ,  $\overline{M}_2 := M(\overline{\theta}_2, \overline{\varphi}_2)$  as well as  $\overline{M}_1^\theta := M^\theta(\overline{\theta}_1, \overline{\varphi}_1)$ ,  $\overline{M}_1^\varphi := M^\varphi(\overline{\theta}_1, \overline{\varphi}_1)$  and analogously for  $\overline{M}_2^\theta, \overline{M}_2^\varphi$ . Finally set

$$G := \langle R(\overline{\alpha})\overline{M}_1 S, w \rangle - \varepsilon \cdot (|\langle R'(\overline{\alpha})\overline{M}_1 S, w \rangle| + |\langle R(\overline{\alpha})\overline{M}_1^\theta S, w \rangle| + |\langle R(\overline{\alpha})\overline{M}_1^\varphi S, w \rangle|) - 9\varepsilon^2/2,$$

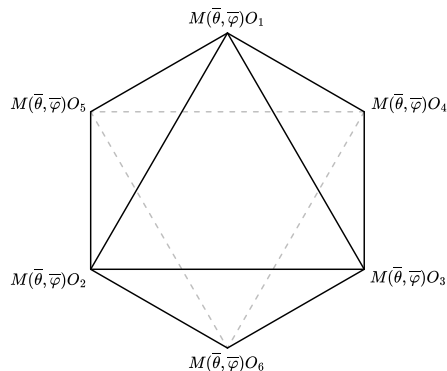
$$H_P := \langle \overline{M}_2 P, w \rangle + \varepsilon \cdot (|\langle \overline{M}_2^\theta P, w \rangle| + |\langle \overline{M}_2^\varphi P, w \rangle|) + 2\varepsilon^2, \quad \text{for } P \in \mathbf{P}.$$

If  $G > \max_{P \in \mathbf{P}} H_P$  then there does not exist a solution to Rupert's condition with

$$(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\overline{\theta}_1 \pm \varepsilon, \overline{\varphi}_1 \pm \varepsilon, \overline{\theta}_2 \pm \varepsilon, \overline{\varphi}_2 \pm \varepsilon, \overline{\alpha} \pm \varepsilon] \subseteq \mathbb{R}^5.$$

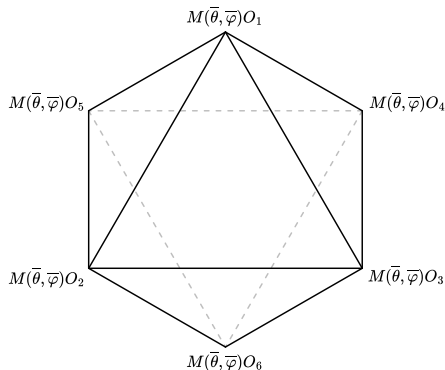
$\varepsilon = 0.08$  from the Example can be replaced by  $\varepsilon = 0.164$

# Motivation for local theorem: $\theta_1 = \theta_2, \varphi_1 = \varphi_2, \alpha = 0$



$$(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) = (\pi/4, \tan^{-1}(\sqrt{2}), \pi/4, \tan^{-1}(\sqrt{2}), 0)$$

# Motivation for local theorem: $\theta_1 = \theta_2, \varphi_1 = \varphi_2, \alpha = 0$



$$A := M(\theta, \varphi)O_1,$$

$$B := M(\theta, \varphi)O_2,$$

$$C := M(\theta, \varphi)O_3$$

**Fact:**

$$\|A\|^2 + \|B\|^2 + \|C\|^2 = 2$$

$\implies$  no local solution  
from this direction

$$(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) = (\pi/4, \tan^{-1}(\sqrt{2}), \pi/4, \tan^{-1}(\sqrt{2}), 0)$$

## Conditions for the three-point method

- $P_1, P_2, P_3$  are all in front of the projection:

$$\langle X(\bar{\theta}, \bar{\varphi}), P_i \rangle > 0.$$

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$$\|M(\bar{\theta}, \bar{\varphi})P_i\| > r.$$



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- The points  $P_1, P_2, P_3$  are locally maximally distant.

## Conditions for the three-point method

- $P_1, P_2, P_3$  are all in front of the projection:

$$\langle X(\bar{\theta}, \bar{\varphi}), P_i \rangle > \sqrt{2}\epsilon.$$

- The origin is inside the projected triangle  $M(\bar{\theta} \pm \epsilon, \bar{\varphi} \pm \epsilon) \cdot \{P_1, P_2, P_3\}$ .

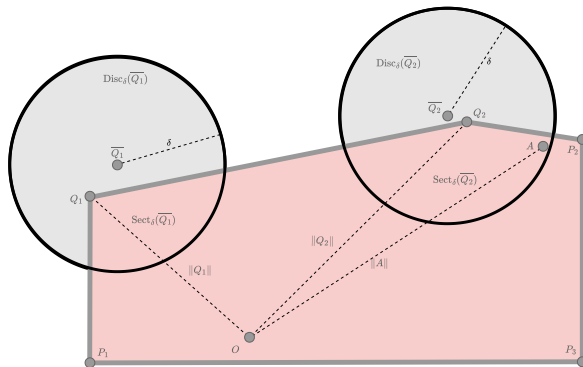
$\epsilon$ -spanning

- The projected points  $M(\bar{\theta}, \bar{\varphi}) \cdot \{P_1, P_2, P_3\}$  are not too close to the origin:

$$\|M(\bar{\theta}, \bar{\varphi})P_i\| > r + \sqrt{2}\epsilon.$$

- The points  $P_1, P_2, P_3$  are  $\epsilon$ -locally maximally distant.

# Locally maximally distant points



# Local Theorem

## Theorem (Local Theorem v0.1)

Let  $\mathbf{P}$  be a convex, pointsymmetric polyhedron with radius 1 and  $P_1, P_2, P_3 \in \mathbf{P}$ . Set  $\bar{X} := X(\bar{\theta}, \bar{\varphi})$ ,  $\bar{M} := M(\bar{\theta}, \bar{\varphi})$ . Assume that

$$\langle \bar{X}, P_i \rangle > \sqrt{2}\varepsilon \quad (\mathbf{A}_\varepsilon)$$

for all  $i = 1, 2, 3$ . Moreover, assume that  $P_1, P_2, P_3$  are  $\varepsilon$ -spanning for  $(\bar{\theta}, \bar{\varphi})$ . Finally, assume that the points  $P_1, P_2, P_3$  are  $\varepsilon$ -LMD. Then there exists no solution to Rupert's problem with  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \pm \varepsilon]$ .

## $\varepsilon$ -spanning points

- $P_1, P_2, P_3$  are all in front of the projection:

$$\langle X(\bar{\theta}, \bar{\varphi}), P_i \rangle > \sqrt{2}\varepsilon.$$

- The origin is inside the projected triangle  $M(\bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon) \cdot \{P_1, P_2, P_3\}$ .

$\varepsilon$ -spanning

### Lemma

Let  $P_1, P_2, P_3$  with  $\|P_1\|, \|P_2\|, \|P_3\| \leq 1$  be  $\varepsilon$ -spanning for  $(\bar{\theta}, \bar{\varphi})$  and let  $\theta, \varphi \in \mathbb{R}$  such that  $|\theta - \bar{\theta}|, |\varphi - \bar{\varphi}| \leq \varepsilon$ . Assume that  $\langle X(\theta, \varphi), P_i \rangle > 0$  for  $i = 1, 2, 3$ . Then

$$X(\theta, \varphi) \in \text{span}^+(P_1, P_2, P_3).$$

## Some more lemmas

### Lemma (Pythagoras)

*For any  $P \in \mathbb{R}^3$  one has  $\|M(\theta, \varphi)P\|^2 = \|P\|^2 - \langle X(\theta, \varphi), P \rangle^2$ .*

### Lemma (Trinity)

*Let  $P_1, P_2, P_3, Y, Z \in \mathbb{R}^3$  with  $\|Y\| = \|Z\|$  and  $Y, Z \in \text{span}^+(P_1, P_2, P_3)$ . Then at least one of the following inequalities does not hold:*

$$\langle P_1, Y \rangle > \langle P_1, Z \rangle,$$

$$\langle P_2, Y \rangle > \langle P_2, Z \rangle,$$

$$\langle P_3, Y \rangle > \langle P_3, Z \rangle.$$

# Proof sketch of the Local Theorem

## Proof sketch.

**1** Assume  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \pm \varepsilon]$ , let

$$M_1 = M(\theta_1, \varphi_1), M_2 = M(\theta_2, \varphi_2), X_1 = X(\theta_1, \varphi_1), X_2 = X(\theta_2, \varphi_2).$$

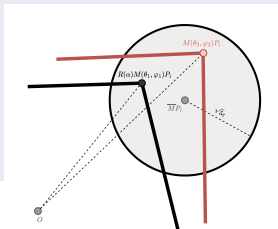
# Proof sketch of the Local Theorem

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$$M_1 = M(\theta_1, \varphi_1), M_2 = M(\theta_2, \varphi_2), X_1 = X(\theta_1, \varphi_1), X_2 = X(\theta_2, \varphi_2).$$

- 2 Rupert's property +  $\varepsilon$ -LMD  $\Rightarrow \|M_2 P_i\| > \|M_1 P_i\|$  for  $i = 1, 2, 3$ .





# Proof sketch of the Local Theorem

## Proof sketch.

- 1 Assume  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \pm \varepsilon]$ , let

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- 2 Rupert's property +  $\varepsilon$ -LMD  $\Rightarrow \|M_2 P_i\| > \|M_1 P_i\|$  for  $i = 1, 2, 3$ .
- 3  $\langle \bar{X}, P_i \rangle > \sqrt{2}\varepsilon \Rightarrow \langle X_1, P_i \rangle, \langle X_2, P_i \rangle > 0$ .

# Proof sketch of the Local Theorem

## Proof sketch.

- 1 Assume  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \bar{\theta} \pm \varepsilon, \bar{\varphi} \pm \varepsilon, \pm \varepsilon]$ , let

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- 4 (2), (3) and (Pythagoras)  $\Rightarrow \langle X_1, P_i \rangle > \langle X_2, P_i \rangle$

# Proof sketch of the Local Theorem

## Proof sketch.

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5  $P_1, P_2, P_3$  are  $\varepsilon$ -spanning (+ Lemma)  $\Rightarrow X_1, X_2 \in \text{span}^+(P_1, P_2, P_3)$

# Proof sketch of the Local Theorem

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3  $\langle \bar{X}, P_i \rangle > \sqrt{2}\varepsilon \Rightarrow \langle X_1, P_i \rangle, \langle X_2, P_i \rangle > 0$ .

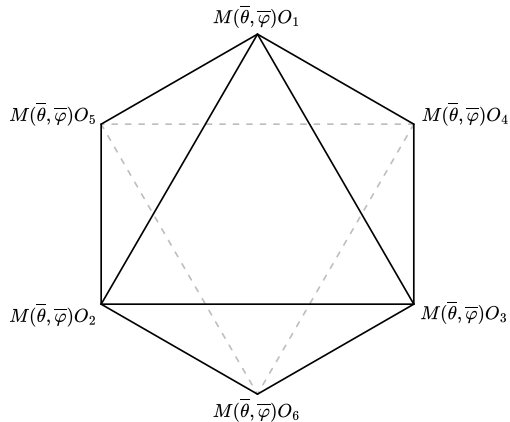
4 (2), (3) and (Pythagoras)  $\Rightarrow \langle X_1, P_i \rangle > \langle X_2, P_i \rangle$

5  $P_1, P_2, P_3$  are  $\varepsilon$ -spanning (+ Lemma)  $\Rightarrow X_1, X_2 \in \text{span}^+(P_1, P_2, P_3)$

6 (4)+(5) + Trinity lemma  $\Rightarrow$  contradiction.



## Example continued



## Theorem (Local Theorem v1.0)

Let  $\mathbf{P}$  be a polyhedron with radius 1 and  $P_1, P_2, P_3, Q_1, Q_2, Q_3 \in \mathbf{P}$ . Assume that  $P_1, P_2, P_3$  and  $Q_1, Q_2, Q_3$  are congruent. Let  $\varepsilon > 0$  and  $\bar{\theta}_1, \bar{\varphi}_1, \bar{\theta}_2, \bar{\varphi}_2, \bar{\alpha} \in \mathbb{R}$ , then set  $\bar{X}_1 := X(\bar{\theta}_1, \bar{\varphi}_1)$ ,  $\bar{X}_2 := X(\bar{\theta}_2, \bar{\varphi}_2)$  as well as  $\bar{M}_1 := M(\bar{\theta}_1, \bar{\varphi}_1)$ ,  $\bar{M}_2 := M(\bar{\theta}_2, \bar{\varphi}_2)$ . Assume that there exist  $\sigma_P, \sigma_Q \in \{0, 1\}$  such that

$$(-1)^{\sigma_P} \langle \bar{X}_1, P_i \rangle > \sqrt{2}\varepsilon \quad \text{and} \quad (-1)^{\sigma_Q} \langle \bar{X}_2, Q_i \rangle > \sqrt{2}\varepsilon, \quad (\text{A}_\varepsilon)$$

for all  $i = 1, 2, 3$ . Moreover, assume that  $P_1, P_2, P_3$  are  $\varepsilon$ -spanning for  $(\bar{\theta}_1, \bar{\varphi}_1)$  and that  $Q_1, Q_2, Q_3$  are  $\varepsilon$ -spanning for  $(\bar{\theta}_2, \bar{\varphi}_2)$ . Finally, assume that for all  $i = 1, 2, 3$  and any  $Q_j \in \mathbf{P} \setminus Q_i$  it holds that

$$\frac{\langle \bar{M}_2 Q_i, \bar{M}_2(Q_i - Q_j) \rangle - 2\varepsilon \|Q_i - Q_j\| \cdot (\sqrt{2} + \varepsilon)}{(\|\bar{M}_2 Q_i\| + \sqrt{2}\varepsilon) \cdot (\|\bar{M}_2(Q_i - Q_j)\| + 2\sqrt{2}\varepsilon)} > \frac{\sqrt{5}\varepsilon + \delta}{r}, \quad (\text{B}_\varepsilon)$$

for some  $r > 0$  such that  $\min_{i=1,2,3} \|\bar{M}_2 Q_i\| > r + \sqrt{2}\varepsilon$  and for some  $\delta \in \mathbb{R}$  with

$$\delta \geq \max_{i=1,2,3} \|R(\bar{\alpha})\bar{M}_1 P_i - \bar{M}_2 Q_i\| / 2.$$

Then there exists no solution to Rupert's problem  $R(\alpha)M(\theta_1, \varphi_1) \mathbf{P} \subset M(\theta_2, \varphi_2) \mathbf{P}^\circ$  with  $(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta}_1 \pm \varepsilon, \bar{\varphi}_1 \pm \varepsilon, \bar{\theta}_2 \pm \varepsilon, \bar{\varphi}_2 \pm \varepsilon, \bar{\alpha} \pm \varepsilon] \subseteq \mathbb{R}^5$ .

# Motivation for rational approximation

 R 4.5.1 · ~ / 

```
> 1 + 10^(-16) > 1
```

```
[1] FALSE
```

```
> |
```

SageMath version 10.0, Release Date: 2023-05-20  
Using Python 3.11.1. Type "help()" for help.

```
sage: 1 + 10^(-16) > 1
```

```
True
```

```
sage: █
```

# Motivation for rational approximation

```
R 4.5.1 · ~/ ↗  
> 1 + 10^(-16) > 1  
[1] FALSE  
> |
```

imprecise but fast

```
SageMath version 10.0, Release Date: 2023-05-20  
Using Python 3.11.1. Type "help()" for help.
```

```
sage: 1 + 10^(-16) > 1  
True  
sage: █
```

exact but slow



# Idea of rational approximation $\kappa := 10^{-10}$

$$\sin_{\mathbb{Q}}(x) := x - \frac{x^3}{3} + \frac{x^5}{5!} \mp \cdots + \frac{x^{25}}{25!},$$

$$\cos_{\mathbb{Q}}(x) := 1 - \frac{x^2}{2} + \frac{x^4}{4!} \mp \cdots + \frac{x^{24}}{24!}.$$

By replacing  $\sin, \cos$  with  $\sin_{\mathbb{Q}}, \cos_{\mathbb{Q}}$  define the functions

$$R_{\mathbb{Q}}(\alpha), R'_{\mathbb{Q}}(\alpha), X_{\mathbb{Q}}(\theta, \varphi), M_{\mathbb{Q}}(\theta, \varphi), M_{\mathbb{Q}}^{\theta}(\theta, \varphi), M_{\mathbb{Q}}^{\varphi}(\theta, \varphi).$$

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By replacing  $\sin, \cos$  with  $\sin_{\mathbb{Q}}, \cos_{\mathbb{Q}}$  define the functions

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## Lemma

Let  $\alpha, \theta, \varphi \in [-4, 4]$ . Then it holds that

$$\|R(\alpha) - R_{\mathbb{Q}}(\alpha)\|, \|R'(\alpha) - R'_{\mathbb{Q}}(\alpha)\|, \|X(\theta, \varphi) - X_{\mathbb{Q}}(\theta, \varphi)\|, \|M(\theta, \varphi) - M_{\mathbb{Q}}(\theta, \varphi)\| \leq \kappa.$$

Moreover,

$$\|R_{\mathbb{Q}}(\alpha)\|, \|R'_{\mathbb{Q}}(\alpha)\|, \|X_{\mathbb{Q}}(\theta, \varphi)\|, \|M_{\mathbb{Q}}(\theta, \varphi)\| \leq 1 + \kappa.$$

# Rational global theorem

## Theorem (Rational Global Theorem)

Let  $\mathbf{P}$  be a pointsymmetric convex polyhedron with radius  $\rho = 1$  and  $\tilde{\mathbf{P}}$  a  $\kappa$ -rational approximation. Let  $\tilde{S} \in \tilde{\mathbf{P}}$ . Further let  $\varepsilon > 0$  and  $\bar{\theta}_1, \bar{\varphi}_1, \bar{\theta}_2, \bar{\varphi}_2, \bar{\alpha} \in \mathbb{Q} \cap [-4, 4]$ . Let  $w \in \mathbb{Q}^2$  be a unit vector. Denote  $\overline{M}_1 := M_{\mathbb{Q}}(\bar{\theta}_1, \bar{\varphi}_1)$ ,  $\overline{M}_2 := M_{\mathbb{Q}}(\bar{\theta}_2, \bar{\varphi}_2)$  as well as  $\overline{M}_1^{\theta} := M_{\mathbb{Q}}^{\theta}(\bar{\theta}_1, \bar{\varphi}_1)$ ,  $\overline{M}_1^{\varphi} := M_{\mathbb{Q}}^{\varphi}(\bar{\theta}_1, \bar{\varphi}_1)$  and analogously for  $\overline{M}_2^{\theta}, \overline{M}_2^{\varphi}$ . Finally set

$$G^{\mathbb{Q}} := \langle R_{\mathbb{Q}}(\bar{\alpha}) \overline{M}_1 \tilde{S}, w \rangle - \varepsilon \cdot (|\langle R'_{\mathbb{Q}}(\bar{\alpha}) \overline{M}_1 \tilde{S}, w \rangle| + |\langle R_{\mathbb{Q}}(\bar{\alpha}) \overline{M}_1^{\theta} \tilde{S}, w \rangle| + |\langle R_{\mathbb{Q}}(\bar{\alpha}) \overline{M}_1^{\varphi} \tilde{S}, w \rangle|) - 9\varepsilon^2/2 - 4\kappa(1 + 3\varepsilon),$$

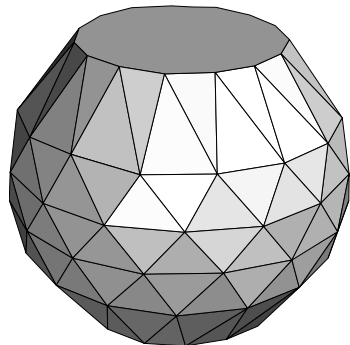
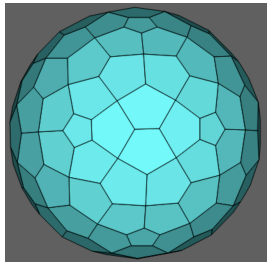
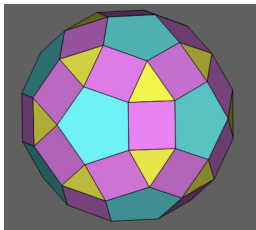
$$H_P^{\mathbb{Q}} := \langle \overline{M}_2 P, w \rangle + \varepsilon \cdot (|\langle \overline{M}_2^{\theta} P, w \rangle| + |\langle \overline{M}_2^{\varphi} P, w \rangle|) + 2\varepsilon^2 + 3\kappa(1 + 2\varepsilon).$$

If  $G^{\mathbb{Q}} > \max_{P \in \tilde{\mathbf{P}}} H_P^{\mathbb{Q}}$  then there does not exist a solution to Rupert's condition with

$$(\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha) \in [\bar{\theta}_1 \pm \varepsilon, \bar{\varphi}_1 \pm \varepsilon, \bar{\theta}_2 \pm \varepsilon, \bar{\varphi}_2 \pm \varepsilon, \bar{\alpha} \pm \varepsilon].$$

# Wishlist for a solid

- 1 Not Rupert
- 2 many symmetries
- 3 no mirror symmetry
- 4 local theorem always applicable
- 5 point symmetry
- 6 not too many vertices



## Definition of the Noperthedron

$$\mathcal{C}_{30} := \left\{ (-1)^\ell R_z \left( \frac{2\pi k}{15} \right) : k = 0, \dots, 14; \ell = 0, 1 \right\}.$$

$$C_1 := \frac{1}{259375205} \begin{pmatrix} 152024884 \\ 0 \\ 210152163 \end{pmatrix}, \quad C_2 := \frac{1}{10^{10}} \begin{pmatrix} 6632738028 \\ 6106948881 \\ 3980949609 \end{pmatrix}, \quad C_3 := \frac{1}{10^{10}} \begin{pmatrix} 8193990033 \\ 5298215096 \\ 1230614493 \end{pmatrix}.$$

Note:  $\|C_1\| = 1$  and  $\frac{98}{100} < \|C_i\| < \frac{99}{100}$  for  $i = 2, 3$ .

### Definition

Define the set of points **NOP**  $\subseteq \mathbb{R}^3$  by the action of  $\mathcal{C}_{30}$  on  $C_1, C_2, C_3$ :

$$\mathbf{NOP} := \mathcal{C}_{30} \cdot C_1 \cup \mathcal{C}_{30} \cdot C_2 \cup \mathcal{C}_{30} \cdot C_3.$$

The *Noperthedron* has 90 vertices. **NOP** is pointsymmetric since  $-\text{Id} \in \mathcal{C}_{30}$ . Symmetries, e.g.:  $M(\theta + 2\pi/15, \varphi) \cdot \mathbf{NOP} = M(\theta, \varphi) \cdot \mathbf{NOP} \Rightarrow \theta_1, \theta_2 \in [0, 2\pi/15)$ .

# Certificate of computer proof

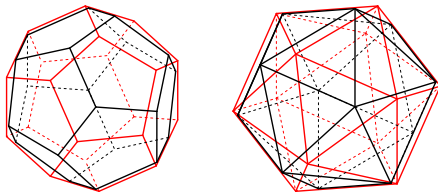
| ID    | T | #C | ID1C  | sp | $\bar{\theta}_{1,\min}$ | $\bar{\theta}_{1,\max}$ | $\bar{\varphi}_{1,\min}$ | $\bar{\varphi}_{1,\max}$ | $\bar{\theta}_{2,\min}$ | $\bar{\theta}_{2,\max}$ | $\bar{\varphi}_{2,\min}$ | $\bar{\varphi}_{2,\max}$ | $\bar{\alpha}_{\min}$ | $\bar{\alpha}_{\max}$ | $P_1$ | $P_2$ | $P_3$ | $Q_1$ | $Q_2$ | $Q_3$ | $r$ | $\sigma_Q$ | $w_x$ | $w_y$  | $w_d$ | $S$ |
|-------|---|----|-------|----|-------------------------|-------------------------|--------------------------|--------------------------|-------------------------|-------------------------|--------------------------|--------------------------|-----------------------|-----------------------|-------|-------|-------|-------|-------|-------|-----|------------|-------|--------|-------|-----|
| 0     | 3 | 4  | 1     | 1  | 0                       | 64.00                   | 0                        | 48.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 1     | 3 | 30 | 5     | 2  | 0                       | 16.00                   | 0                        | 48.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 2     | 3 | 30 | 46.67 | 2  | 16.00                   | 32.00                   | 0                        | 48.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 3     | 3 | 30 | 94.77 | 2  | 32.00                   | 48.00                   | 0                        | 48.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 4     | 3 | 30 | 14.51 | 2  | 48.00                   | 64.00                   | 0                        | 48.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 5     | 3 | 4  | 35    | 3  | 0                       | 16.00                   | 0                        | 16.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 6     | 3 | 4  | 70.31 | 3  | 0                       | 16.00                   | 16.00                    | 32.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| 7     | 3 | 4  | 10.67 | 3  | 0                       | 16.00                   | 32.00                    | 48.00                    | 0                       | 64.00                   | 0                        | 24.00                    | -24.00                | 24.00                 |       |       |       |       |       |       |     |            |       |        |       |     |
| ⋮     | ⋮ | ⋮  | ⋮     | ⋮  | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                     | ⋮                     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮   | ⋮          | ⋮     | ⋮      | ⋮     | ⋮   |
| ⋮     | ⋮ | ⋮  | ⋮     | ⋮  | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                     | ⋮                     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮   | ⋮          | ⋮     | ⋮      | ⋮     | ⋮   |
| 91    | 1 |    |       |    | 0                       | 80.00                   | 0                        | 80.00                    | 80.00                   | 16.00                   | 80.60                    | 16.20                    | -23.40                | -22.80                |       |       |       |       |       |       |     |            | 53.73 | 15.64  | 16.45 | 39  |
| 92    | 1 |    |       |    | 0                       | 80.00                   | 80.00                    | 16.00                    | 0                       | 80.00                   | 0                        | 80.60                    | -24.00                | -23.40                |       |       |       |       |       |       |     |            | 98.92 | 35.15  | 10.33 | 37  |
| ⋮     | ⋮ | ⋮  | ⋮     | ⋮  | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                     | ⋮                     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮   | ⋮          | ⋮     | ⋮      | ⋮     | ⋮   |
| 245   | 2 |    |       |    | 0                       | 20.00                   | 0                        | 20.00                    | 0                       | 20.00                   | 0                        | 20.40                    | -22.20                | -22.80                | 30    | 31    | 38    | 79    | 80    | 87    | -69 | 1          |       |        |       |     |
| 246   | 1 |    |       |    | 0                       | 20.00                   | 0                        | 20.00                    | 0                       | 20.00                   | 20.40                    | 40.80                    | -23.60                | -22.20                |       |       |       |       |       |       |     |            | 71.05 | 19.88  | 20.37 | 39  |
| 247   | 1 |    |       |    | 0                       | 20.00                   | 0                        | 20.00                    | 0                       | 20.00                   | 20.40                    | 40.80                    | -22.20                | -22.80                |       |       |       |       |       |       |     |            | 71.05 | 19.88  | 20.37 | 39  |
| 248   | 2 |    |       |    | 0                       | 20.00                   | 0                        | 20.00                    | 20.00                   | 40.00                   | 0                        | 20.40                    | -23.60                | -22.20                | 30    | 31    | 38    | 79    | 80    | 87    | -69 | 1          |       |        |       |     |
| 249   | 2 |    |       |    | 0                       | 20.00                   | 0                        | 20.00                    | 20.00                   | 40.00                   | 0                        | 20.40                    | -22.20                | -22.80                | 30    | 31    | 38    | 79    | 80    | 87    | -69 | 1          |       |        |       |     |
| ⋮     | ⋮ | ⋮  | ⋮     | ⋮  | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                       | ⋮                       | ⋮                        | ⋮                        | ⋮                     | ⋮                     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮     | ⋮   | ⋮          | ⋮     | ⋮      | ⋮     | ⋮   |
| 18.44 | 1 |    |       |    | 48.00                   | 64.00                   | 46.00                    | 48.00                    | 48.00                   | 64.00                   | 22.80                    | 24.00                    | 22.80                 | 24.00                 |       |       |       |       |       |       |     |            | 33.40 | -14.51 | 14.49 | 78  |

# Computer proof statistics

- $\approx 18.000.000$  global theorem applications
- $\approx 600.000$  local theorem applications
- $\approx 3\text{GB}$  uncompressed certificate ( $\approx 150\text{MB}$  compressed)
- $\approx 10\text{h}$  for creation of table (using floating points in R)
- $\approx 30\text{h}$  for verification in SageMath

## Conclusion and open question

- What about the remaining 3 Archimedean solids? In particular the Rhombicosidodecahedron?
- Is there a way to prove that a solid does not have Rupert's property without a huge case distinction? In particular, are there other ways to disprove the existence of local solutions?
- How to prove the conjectured optimal solutions?
- What is the link to duality?





## Bonus: Nieuwland's number

### Fact (Nieuwland, 1816)

It is possible to cut a hole inside the **unit cube** such that a **cube with side length less than**  $3\sqrt{2}/4 \approx 1.0606$  can pass through this hole.

- The largest number  $\nu \in \mathbb{R}$  such that  $\nu\mathbf{P}$  passes through some hole inside  $\mathbf{P}$  is called *Nieuwland number of  $\mathbf{P}$* .
- For all solids it holds that  $\nu \geq 1$ .
- $\mathbf{P}$  is Rupert  $\iff \nu(\mathbf{P}) > 1$ .
- $\nu$  of the Cube is exactly  $3\sqrt{2}/4$  [Nieuwland, 1816].

# Sufficient condition of LMD

## Lemma

Let  $\mathcal{P}$  be a convex polygon and  $Q \in \mathcal{P}$ . Let  $\bar{Q} \in \mathbb{R}^2$  with  $\|Q - \bar{Q}\| < \delta$  for some  $\delta > 0$ . Assume that for some  $r > 0$  such that  $\|Q\| > r$  it holds that

$$\frac{\langle Q, Q - P_j \rangle}{\|Q\| \|Q - P_j\|} \geq \frac{\delta}{r},$$

for all vertices  $P_j \in \mathcal{P} \setminus Q$ . Then  $Q \in \mathcal{P}$  is  $\delta$ -LMD( $\bar{Q}$ ).

## Proof sketch.

Assume  $A \in \text{Sect}_\delta(\bar{Q}) = \text{Disc}_\delta(\bar{Q}) \cap \mathcal{P}^\circ$ , use  $\cos(\angle(O, Q, P_j)) = \frac{\langle Q, Q - P_j \rangle}{\|Q\| \|Q - P_j\|} \geq \frac{\delta}{r}$  to conclude that  $\cos(\angle(O, Q, A)) \geq \delta/r$ . Therefore,

$$\|A\|^2 - \|Q\|^2 = \|Q - A\| \cdot (\|Q - A\| - 2\|Q\| \cos(\angle(O, Q, A))) < 0. \quad \square$$

## Bonus: Proof of main lemma for global theorem

$$R_x(\alpha) := \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix}, R_y(\alpha) := \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix}, R_z(\alpha) := \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$X(\theta, \varphi)^t = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \cdot R_y(\varphi) \cdot R_z(-\theta), \quad M(\theta, \varphi) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \cdot R_y(\varphi) \cdot R_z(-\theta)$$

$$R(\alpha)M(\theta, \varphi) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \cdot R_z(\alpha) \cdot R_y(\varphi) \cdot R_z(-\theta).$$

### Lemma

For any  $\alpha, \beta \in \mathbb{R}$  one has  $\|R_x(\alpha)R_y(\beta) - \text{Id}\| \leq \sqrt{\alpha^2 + \beta^2}$ .

Sketch of proof: Write  $R_x(\alpha)R_y(\beta) = UR_x(\Phi)U^{-1}$  and take trace to obtain:

$$\cos(\alpha) + \cos(\beta) + \cos(\alpha)\cos(\beta) = 1 + 2\cos(\Phi).$$

Jensen on  $f(t) = \cos(\sqrt{t})$  shows  $\text{LHS} \geq 1 + 2\cos(\sqrt{\alpha^2 + \beta^2})$ , thus  $|\Phi| \leq \sqrt{\alpha^2 + \beta^2}$ .